

CSA0609-DESIGN AND ANALYSIS OF ALGORITHMS

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Exp-05: Out of Boundary Paths

Aim:

To find the number of paths by which a ball can move out of the boundary of a grid within a given number of moves

Algorithm:

1. Read the number of rows, columns, maximum moves, and starting position.
2. If the current position is outside the grid, return 1.
3. If the maximum number of moves becomes zero, return 0.
4. Move the ball in four directions: up, down, left, and right.
5. Recursively calculate the number of paths that go outside the boundary.
6. Add the paths from all four directions.
7. Display the number of paths and complexity.

Source Code:

```
def paths(m, n, moves, row, col):  
    if row < 0 or row >= m or col < 0 or col >= n:  
        return 1  
  
    if moves == 0:  
        return 0  
  
    return (paths(m, n, moves - 1, row - 1, col) +  
            paths(m, n, moves - 1, row + 1, col) +  
            paths(m, n, moves - 1, row, col - 1) +
```

```
paths(m, n, moves - 1, row, col + 1))
```

```
m = int(input("Enter number of rows: "))
```

```
n = int(input("Enter number of columns: "))
```

```
moves = int(input("Enter maximum moves: "))
```

```
row = int(input("Enter starting row: "))
```

```
col = int(input("Enter starting column: "))
```

```
result = paths(m, n, moves, row, col)
```

```
print("Number of Out of Boundary Paths:", result)
```

```
print("Time Complexity:  $O(4^N)$ ")
```

```
print("Space Complexity:  $O(N)$ ")
```

Output:

```
Enter number of rows: 2
Enter number of columns: 2
Enter maximum moves: 2
Enter starting row: 0
Enter starting column: 0
Number of Out of Boundary Paths: 6
Time Complexity:  $O(4^N)$ 
Space Complexity:  $O(N)$ 
```

Result:

The number of paths that move the ball outside the boundary was successfully calculated.